

$$\begin{aligned}
& \mathbf{J} \times \mathbf{H} \\
& \mathbf{E} \mathbf{H} \\
& \mathbf{j} \\
& \mathbf{S} \\
& \mathbf{L} \\
& (r,\phi,z) \\
& e^{i\ell\phi} = \\
& \tilde{L} \hbar \\
& L_z \\
& \ell \hbar \\
& \partial E_x/\partial x \\
& \partial E_y/\partial y \\
& \nabla \cdot \\
& \mathbf{E} = \\
& \frac{\partial E_x}{\partial x} + \\
& \frac{\partial E_y}{\partial y} + \\
& \frac{\partial E_z}{\partial z} = \\
& 0 \\
& E_z \\
& E_x, E_y \\
& \frac{s \pm 1}{s \pm 1} \\
& \frac{s \pm 1}{s \pm 1} = \\
& \frac{in}{\mathbf{E}_0}(r,z)(\hat{x} + \\
& is\hat{y})e^{ikz} \\
& \frac{in}{\alpha} \\
& (\hat{x} + \\
& is\hat{y}) = \\
& (\cos \phi \hat{\rho} - \\
& \sin \phi \hat{\phi}) + \\
& is(\sin \phi \hat{\rho} + \\
& \cos \phi \hat{\phi}) \\
& \theta \\
& \hat{\phi} \\
& \hat{z} \\
& \hat{\rho} \\
& \hat{z} \\
& \hat{\rho} \\
& \alpha \\
& \cos \phi + \\
& is \sin \phi \\
& \cos \phi + \\
& is \sin \phi = \\
& e^{is\phi} \\
& \theta \\
& \phi_p \\
& E_z \\
& \tilde{\alpha} \\
& \int_0^{2\pi} (\cos \phi_p + \\
& is \sin \phi_p) e^{ik\rho \cos(\phi_p - \phi)} d\phi_p \\
& \int_0^{2\pi} e^{is\phi_p} e^{ik\rho \cos(\phi_p - \phi)} d\phi_p = \\
& e^{is\phi} \int_0^{2\pi} e^{is\alpha} e^{ik\rho \cos \alpha} d\alpha \\
& e^{is\phi} \\
& J_1(k\rho \sin \theta) \\
& \tilde{0} = \\
& \hat{0} = \\
& \hat{z} \\
& (\rho, \phi, z) = \\
& -iAe^{is\phi} I_1(\rho, z) \\
& A \\
& I_1(\rho, z) \\
& NA = \\
& n \sin \alpha \\
& {}_1(\rho, z) = \\
& \int_0^\alpha \cos^{1/2} \theta \sin^2 \theta P(\theta) J_1(k\rho \sin \theta) e^{ikz \cos \theta} d\theta \\
& \hat{\theta} \\
& \hat{\alpha} \\
& \hat{P}(\theta) \\
& J_1 \\
& E_z \\
& e^{is\phi} \\
& \ell \equiv \\
& s \equiv \\
& \frac{s+1}{s+1} \\
& \frac{s+1}{s+1} \\
& \ell \equiv \\
& \frac{s+1}{s+1} \\
& \hat{z} \\
& \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + 2ik \frac{\partial}{\partial z} \right) u(r, \phi, z) = \\
& 0 \\
& \exp(i\ell\phi) \\
& \frac{2\pi}{\ell} \\
& \hat{\ell} \\
& \hat{0} = 1 + 2
\end{aligned}$$